PharmAchieve

DEMENTIA

Qualifying Exam Preparatory Course

COVERED IN THIS PRESENTATION…

Providing Care: Clinical Care

Providing Care: Distribution

Knowledge & Expertise

Communication & Collaboration

Leadership & Stewardship

Professionalism

LEGEND

ACh	Acetylcholine
AChEI	Acetylcholinesterase Inhibitor
AD	Alzheimer's Disease
ADAS-Cog	Alzheimer's Disease Assessment Scale-Cognitive Subscale
ADD	Alzheimer's Disease Dementia
ADL	Activities of Daily Living
ADR	Adverse Drug Reaction
BID	Twice Daily
BP	Blood Pressure
BPSD	Behavioural and Psychological Symptoms of Dementia
CBT	Cognitive Behavioural Therapy
CrCl	Creatinine Clearance
CV	Cardiovascular
CYP	Cytochrome P450
DASH	Dietary Approaches to Stop Hypertension
EPS	Extrapyramidal Symptoms
FTD	Frontotemporal Dementia
H2RA	Histamine-2 Receptor Antagonist
IADL	Instrumental Activities of Daily Living
IM	Intramuscular
LBD	Lewy Body Dementia
MIND	Mediterranean-DASH Intervention for Neurodegenerative Delay
MMSE	Mini-Mental State Examination
MOA	Mechanism of Action
MoCA	Montreal Cognitive Assessment
NMDA	N-Methyl-D-aspartic acid or N-Methyl-D-aspartate
N/V/D	Nausea/Vomiting/Diarrhea
PDD	Parkinson's Disease Dementia
PO	Per Os (Oral)
PPI	Proton Pump Inhibitor
PRN	Pro Re Nata (As Required)
PSW	Personal Support Worker
SC	Subcutaneous
SSRI	Selective Serotonin Reuptake Inhibitor
SSS	Sick Sinus Syndrome
TCA	Tricyclic Antidepressant
VD	Vascular Dementia

DEFINITION

Dementia is a set of symptoms linked to brain function, characterized by a decline in cognitive function resulting in diminished ability to perform daily activities

Can affect:

- Memory
- Awareness of person, place, and time
- Language
- Basic math skills
- Judgement
- Planning

It is a chronic, progressive, deteriorating condition

Behavioural and psychological symptoms of dementia (BPSD): aggression, agitation, anxiety, apathy, delusions, depression, hallucinations, paranoia, sleep disturbances, wandering

PATHOPHYSIOLOGY

Controversy exists regarding the aβ*56 amyloid protein and its role in Alzheimer's Disease. Despite this, the amyloid theory is still a viable one and is being researched further.

ALZHEIMER'S DISEASE = **Beta Amyloid Plaques** *Dense protein & cellular material deposits* + **Tau Protein Clumps** *Neurofibrillary tangles*

Resulting in ↓ACh + ↑glutamate

AD progresses over a course of ~10 years

MILD → MODERATE → SEVERE

Mild: Cognitive decline begins
- Memory loss
- Losing possessions
- Difficulty with finances
- Impairment in instrumental activities of daily living (e.g. driving, cooking)

Moderate: Personality changes peak
- Getting lost and wandering
- Anxiety/depression
- Forgetting personal history
- Impairment in personal activities of daily living (e.g. bathing, dressing)

Severe: Motor changes peak
- Agitation/aggression
- Hallucinations/delusions
- Sleep disturbances
- Personal activities of daily living cannot be done alone

TYPES OF DEMENTIA

Alzheimer's Disease

- Gradual onset and progressive
- Typically begins with selective memory impairment and cognitive decline (language, visuospatial function) and progresses to motor impairments (apraxia, seizures, extrapyramidal symptoms)
- Protein buildup and plaque development → neuronal degeneration

Frontotemporal Lobe Dementia

- More progressive and common cause of early-onset dementia
- Socially inappropriate behaviours, disinhibition, aphasia
- Atrophy in frontal and/or temporal lobes

TYPES OF DEMENTIA

Vascular Dementia

- More sudden and can be a stepwise decline after a cerebrovascular event
- Acute and/or chronic reduction of blood flow to the brain
- Any dementia caused by cerebrovascular disease or impaired cerebral blood flow (stroke is a common cause, can be other cerebrovascular events)

Lewy Body/Parkinson's Dementia

- Gradual and progressive
- Visual hallucinations, parkinsonism, cognitive fluctuations, rapid eye movement (REM) sleep behavior disorder (RBD), dysautonomia and neuroleptic sensitivity may occur
- Development of Lewy bodies in nerve cells

CLINICAL PRESENTATION

Cognitive Symptoms
(present throughout the illness)

1. Memory Loss
2. Aphasia
   - Impairment of language ability
3. Disorientation
4. Agnosia
   - Inability to recognize things/people
5. Apraxia
   - Unable to perform a task/movement when instructed despite understanding and wanting to do the task

Non-Cognitive Symptoms
(behavioural symptoms, less predictable)

1. Depression
2. Psychotic symptoms
   - Hallucinations and delusions
3. Behavioural disturbances/Responsive behaviours
   - E.g. aggression, uncooperativeness, wandering, sundowning

Functional Symptoms
(appears towards end of illness)

1. Inability to care for self
   - Dressing/bathing/toileting/eating

DEMENTIA VS DELIRIUM

	Dementia	Delirium
Onset	Gradual	Rapid
Course	Progressive	Fluctuates
Duration	Long-term	Short-term
Attention	Normal (affected much later)	Fluctuates
Orientation	Initially intact	Periodically impaired
Reversibility	No	Yes

RISK FACTORS

Modifiable:

1. Less education (to a maximum of 11–12 years of age)
2. Social isolation
3. Late-life depression
4. Physical inactivity
5. Hearing impairment
6. Vascular risk factors (e.g. hypertension, diabetes mellitus, obesity, smoking)
7. Air pollution
8. Traumatic brain injury
9. Alcohol consumption

Non-modifiable:

- Age >65 years
- Genetics
- Females > males (evidence is not strong)

DRUG CONTRIBUTORS

Many drug-induced causes of dementia are due to anticholinergic effects of medications

- Antidepressants: TCAs, paroxetine
- Antiemetics: dimenhydrinate, ondansetron, scopolamine, prochlorperazine, promethazine, chlorpromazine
- Antiepileptics: carbamazepine, oxcarbazepine
- Antihistamines: First generation mainly, cetirizine most commonly in second generation
- Antimuscarinics (class effect): darifenacin, fesoterodine, oxybutynin, solifenacin, trospium, tolterodine
- Antipsychotics (class effect): First generation more so than second generation (still occurs)
- Antiparkinsonian: amantadine, benztropine, trihexyphenidyl, procyclidine
- Antispasmodics: atropine, hyoscyamine
- Antiulcer: H2RAs and PPIs have conflicting evidence
- Gabapentin: New large study shows increased risk, especially in non-elderly adults (research ongoing)
- Hypnotics: zopiclone, zolpidem, eszopiclone, benzodiazepines (long-acting agents especially)
- Muscle relaxants: cyclobenzaprine, methocarbamol, tizanidine

STAGES OF DEMENTIA

FYI

STAGE	CHARACTERISTICS	CORRESPONDING FAST RATING
Preclinical	Subjective complaints accompanied by very mild objective cognitive decline; functioning is unimpaired. This stage has considerable overlap with normal aging and may or may not progress to dementia	3
Mild	Impaired instrumental activities of daily living (IADL), e.g. driving, medication use, finances, cooking, shopping, use of telephone/electronic devices and housekeeping	4
Moderate	In addition to impaired IADL, some personal activities of daily living (ADL) such as bathing, grooming, dressing, toileting and feeding require assistance or prompting	5
Severe	Personal ADL cannot be done independently, even with prompting	6
Terminal	Patients must be fed and become immobile and non-verbal	7

GOALS OF THERAPY

- ✓ Alter the natural disease progression
- ✓ Stabilize and preserve cognitive, social, and physical functioning
- ✓ Minimize adverse effects related to medications
- ✓ Educate about the progression of the disease and alleviate caregiver anxiety and burden

DISEASE PREVENTION TIPS

1. Be physically active
2. Avoid smoking and excessive alcohol consumption
3. Track your numbers. Keep your blood pressure, cholesterol, blood sugar, and weight within recommended ranges
4. Stay socially connected with your family, friends and community
5. Make healthy food choices. The Mediterranean diet and MIND diets are recommended
6. Reduce stress
7. Obtain 7-8 hours of sleep daily, avoid <5 hours of sleep
8. Challenge your brain by trying something new, playing games, or learning a new language
9. Take care of your hearing. Avoid being continuously exposed to loud sounds and wear a hearing aid if hearing does become a problem
10. Lower your risk of falls. Consider installing handrails on all stairs and grab bars in bathrooms.
11. Reduce your exposure to air pollution, such as exhaust from heavy traffic

NON-PHARMACOLOGICAL MEASURES

1st line for managing behavioural symptoms (e.g. agitation)

- Education of disease progression and quality of life issues
- Educate caregivers on strategies such as distraction, redirection, avoiding confrontation, reassurance, and providing stimulation and structure
- Daily rituals
- Exercise: group or individual – no specific duration or intensity
- Mild-moderate dementia – group cognitive stimulation therapy
- Mild dementia with depression/anxiety: structured psychological interventions (e.g. CBT, supportive counselling)
- Patients with dementia with agitation: recreation therapy, exercise, outdoor activities, group activities, massage, music therapy, sensory stimulation and aromatherapy are recommended
- Inspect living environment and address hazards: to reduce risk of falls
- Counsel patients against driving after progressing beyond the initial stages of the disease
- Power of attorney and advance health directives before disease causes incompetence

PHARMACOLOGICAL MEASURES - OVERVIEW

	ADD	VD	LBD and PDD	FTD	Responsive behaviours
Cholinesterase inhibitors	Yes	Maybe (donepezil?)	Yes (rivastigmine and donepezil)	No	Yes*
NMDA receptor antagonists	Yes	Maybe (limited evidence)	Less evidence, maybe for LBD	No	Yes*
Antipsychotics	-	-	Avoid if possible	Avoid if possible	Avoid if possible
Antidepressants	-	-	-	SSRIs, less evidence	Yes
Benzodiazepines	-	-	-	-	Avoid if possible

* Cholinesterase inhibitors and memantine primarily slow cognitive and functional decline in dementia (*indirect* effect on responsive behaviours)

MANAGEMENT OF NEWLY DIAGNOSED DEMENTIA

Newly Diagnosed Dementia (reversible causes ruled out)

- **Alzheimer Dementia** / **Mixed Alzheimer Vascular Dementia**
  - Mild-moderate: donepezil, rivastigmine or galantamine
  - Severe: donepezil; consider adding memantine
- **Vascular Dementia**
  - Consider donepezil (low evidence)
- **Lewy-Body Dementia** / **Parkinson Disease Dementia**
  - Mild-moderate: donepezil or rivastigmine; consider galantamine if others not tolerated
  - Severe: trial a cholinesterase inhibitor; consider adding memantine for LBD (no evidence for PDD)
- **Frontotemporal Dementia**
  - Do not use cholinesterase inhibitors or memantine

\+

- Non-pharmacologic strategies (exercise, cognitive stimulation, social engagement)
- Assessment of driving safety
- Caregiver support
- Advance care planning

LEGEND

LBD: Lewy-body dementia
PDD: Parkinson disease dementia

ACETYLCHOLINESTERASE INHIBITORS

- MOA: Blocks breakdown of ACh → increase in ACh levels in synaptic cells → increased function of neural cells
- 1st line for most types of dementia → slows progression of dementia/role in disease stabilization
- Most evidence in mild-moderate disease except for donepezil (indicated for all severities)
  - All AChEI shown to have similar efficacy
- If symptoms worsen within first year of initiation (minimum 6 months)
  - Ensure patient is on maximum dose, then consider switching to another AChEI
- If patient has been on the cholinesterase inhibitors for several years and has a lack of benefit
  - Consider adding memantine as adjunctive therapy to donepezil

Adverse effects	Considerations	Comments

Urinary incontinence, N/V/D (take with food), fatigue, muscle cramps, anorexia, headache, insomnia, vivid dreams (take in AM), tremor, bradycardia. Small risk of QT interval prolongation	• Donepezil: Is best tolerated, metabolized by CYP3A4, 2D6
	  • Has a long half-life (~70 hours)
	• Rivastigmine: Patch available for patients unable to swallow and has better tolerability than PO
	  • NOT metabolized by CYP pathway
	  • Short half-life (1.5-3 hours depending on route)
	• Galantamine: Metabolized by CYP3A4, 2D6
	  • Renally adjusted
	  • Half-life: 6-8 hours
		• Start low and adjust dose every 4 weeks
		• Efficacy: months

WHEN SHOULD ACETYLCHOLINESTERASE INHIBITORS BE DISCONTINUED?

Doses should be tapered before stopping the agent

- Patient did not experience any clinically meaningful benefit after trying different acetylcholinesterase inhibitors for a suitable period
  - ≥6 months (unless intolerable side effects prior)
- Patient is non-adherent and continued prescribing is without merit
- Patient's rate of cognitive, functional, or behavioural decline is greater on treatment
- Patient experiences intolerable side effects related to therapy
- Comorbidities of patient make continued use of agent unacceptably risky or futile
  - Examples: Terminal illness, bradycardia, profound QT prolongation
- Patient's dementia progresses to a stage where there would be no benefit from continued therapy (severe stage or terminal illness)

NMDA RECEPTOR ANTAGONIST - MEMANTINE

- MOA: Blocks glutamate induced neuronal excitotoxicity (a process implicated as a final common pathway in neuronal death)
- Minimal benefit as monotherapy and should be used as adjunctive therapy to donepezil when indicated for moderate to severe dementia
  - Monotherapy should be considered in patients unable to use cholinesterase inhibitors

Adverse effects	Considerations	Comments
Dizziness, drowsiness, confusion, insomnia, headache, restlessness, ↑BP, cough	• Caution in history of seizures, clinical cardiovascular disease, bradycardia • Avoid in angle-closure glaucoma, SSS, left bundle branch block	• Renal adjustment for CrCl <30 mL/min • Metabolism not affected by CYP pathway • In addition to slowing decline, may provide behavioral benefits

EFFICACY OF CHOLINESTERASE INHIBITORS AND MEMANTINE

FYI

- Average improvement of ADAS-Cog score from use of any cholinesterase inhibitor was 2-3 points. A 3-point change is considered clinically relevant.

	ADAS-Cog	MMSE	MoCA
Score	11 categories with a total score out of 70, higher score means more impairment Score >18 = cognitive impairment	15 parts with a total score of 30, higher score means less impairment Score ≥26 = no impairment	30 question test with a score out of 30, higher score means less impairment Score of ≥26 = no impairment
Advantages	Comprehensive, most research in cognitive decline utilizes this scale	Simple and reliable, plays an important role in clinical practice for screening. More sensitive for severe conditions and assessing change	Simple and reliable, plays an important role in clinical practice for screening. More sensitive than MMSE for detecting mild impairment.
Disadvantages	Cumbersome and not practical in clinical practice	Lacks the same level of sensitivity as the ADAS-Cog Age, education, and ethnicity can affect the score	Lacks the same level of sensitivity as the ADAS-Cog

PHARMACOLOGICAL MEASURES - RESPONSIVE BEHAVIOURS

Antidepressants: SSRIs (citalopram)

- Citalopram is recommended for moderate agitation
- TCAs are recommended against
- Antidepressants are recommended for moderate-severe depression

Benzodiazepines

- For severe agitation associated with imminent risk of harm; short-term use
- Use shorter-acting agents without active metabolites – e.g. lorazepam
- Risk of oversedation, falls, worsening cognition

Other agents:

- Carbamazepine can be considered if agitation symptoms are refractory to other agents
- Valproic against, divalproex: recommended against
- Trazodone: recommended against
- Others e.g. pindolol, lithium, buspirone, ginkgo biloba (herbal): Not recommended due to lack of evidence. Ginkgo may increase bleeding risk

PHARMACOLOGICAL MEASURES - RESPONSIVE BEHAVIOURS

Antipsychotics: olanzapine, quetiapine, risperidone, aripiprazole, brexpiprazole

- Should be avoided if possible; non-pharmacological measures should be tried first
- Short-term use of second-generation antipsychotics may be considered in severe agitation when there is risk of harm to self or others
  - If used, aripiprazole, brexpiprazole and risperidone are preferred for severe agitation
  - Quetiapine can be trialled as 2nd line if above agents are ineffective
- May worsen psychosis in LBD
- Olanzapine is NOT recommended except for potential short-term emergency treatment of severe acute agitation
- Consider deprescribing if no history of severe agitation, psychosis or other appropriate indication
- Antipsychotic use has been associated with increased risk of mortality and cerebrovascular events (e.g. stroke) in the elderly population

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	**Degree of Change**	**Timeframe**
Cognitive Symptoms	Improved	Early improvement: 4-6 weeks Improvement: 3-6 months
Decline in MMSE score	Less than 2 points	Each year

Safety Endpoints		
Parameter	**Degree of Change**	**Timeframe**
Specific Drug ADRs (vivid dreams, N/V, insomnia, dizziness, bradycardia	Prevent or minimize *May ↓dose to maximum tolerated dose or change to another AChEI	Duration of therapy

*Follow up in 2 weeks after initiating medication or a dose change

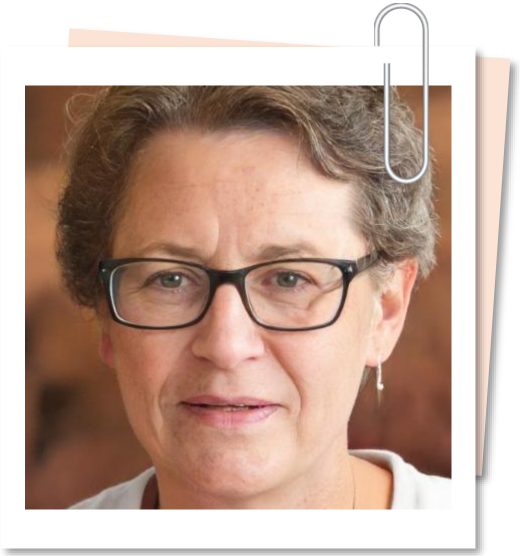
TAKEAWAY

- Dementia is a set of symptoms and occurs gradually and progressively whereas delirium is rapid and fluctuates
- Non-pharm interventions are first-line for managing behavioural symptoms
- Cholinesterase inhibitors are first-line pharmacologic options except in frontotemporal dementia, where they are not effective
- NMDA receptor antagonists should be used as adjunctive therapy to donepezil or if the patient cannot tolerate cholinesterase inhibitors
- Citalopram has the most evidence amongst antidepressants for managing agitation
- Antipsychotics should only be used to help manage responsive behaviours if they have failed other measures and if the patient is at risk of harming themselves or others
  - Antipsychotics can increase risk of mortality and cerebrovascular events in the elderly
- Routinely review psychotropics (antipsychotics, benzodiazepines, antidepressants) and consider deprescribing if appropriate

CASE 1

PT is an 82-year-old female who has been experiencing worsening symptoms of dementia. Initially, she had difficulty finding the words she wanted to use, and she was unable to consistently recognize friends and acquaintances. Over the past few weeks, she has become increasingly agitated, especially at night, and requires more help with personal hygiene, eating, and dressing. PT has a medical history of atrial fibrillation, Alzheimer's dementia, osteoporosis, and remote breast cancer.

PT is taking apixaban 2.5 mg PO BID, metoprolol 50 mg PO BID, donepezil 10 mg PO daily, denosumab 60 mg SC q6months, vitamin D 1000 units PO daily, calcium carbonate 1250 mg PO daily, turmeric 1 tab PO daily, and ginkgo 1 tab PO PRN agitation. She is a widow and lives alone at home with PSW support. She quit smoking 15 years ago. She has a documented allergy with hives to penicillin.



CASE 1 - QUESTIONS

1. Which of the following would you recommend regarding PT's current therapy?
   a) Immediately discontinue donepezil due to lack of efficacy
   b) Reduce dose of donepezil due to increased risk of bradycardia with metoprolol
   c) Discontinue gingko due to increased risk of bleeding with apixaban
   d) Split the dose of donepezil BID to help with PT's evening symptoms

2. As PT's dementia progresses, she is having increasing difficulties with swallowing. A speech language pathologist has suggested administering medications via a non-oral route if possible. What do you recommend regarding PT's dementia therapy?
   a) Switch donepezil to lorazepam sublingual tablets
   b) Switch donepezil to aripiprazole IM
   c) Switch donepezil to rivastigmine patch
   d) Switch donepezil tablets to donepezil patch

CASE 1 - QUESTIONS

3. Which of the following non-pharmacological strategies would be most beneficial for PT to adopt?
   a) Group music therapy
   b) Cognitive behavioural therapy
   c) Cognitive training through completing puzzles
   d) Adopting the DASH diet

4. PT brings a prescription for rivastigmine 9 mg patch, applied once daily for 3 months with 3 refills. What is the total quantity of patches authorized by the prescriber?
   a) 90
   b) 180
   c) 360
   d) 270

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CHANGE LOG

August 2025
- Grammar, spelling and formatting changes made throughout
- Slides 5-7: Reordered and split for clarity
- Slide 11: Added gabapentin based on new study
- Slide 22: Added trazodone can be used in late day confusion
- Slide 24: Updated memantine to be adjunct to donepezil
- Slide 26: Adjusted choices for Question 1

November 2025
- Slide 16: Added note that cholinesterase inhibitors and memantine indirectly help with responsive behaviours
- Slides 22 and 23: Split pharmacological measures to responsive behaviours into 2 slides, and added information to antipsychotic use on slide 23
- Slide 25: Updated takeaway

March 2026
- Slide 2: Updated NAPRA competencies
- Slides 6 & 7: Changes title to *Types* of Dementia
- Slide 8: Added responsive behaviours, and sundowning
- Slide 10: Simplified risk factors
- Slide 16: Updated wording in table
- Slide 22: Updated recommendations, simplified info by removing agents not directly mentioned in guidelines
- Slide 23: Minor updates
- Slide 28: Removed question 2 (case now has only 4 questions)